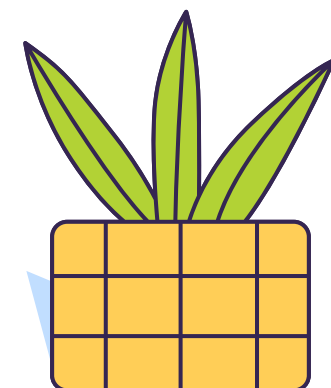


MayoHap Methodology

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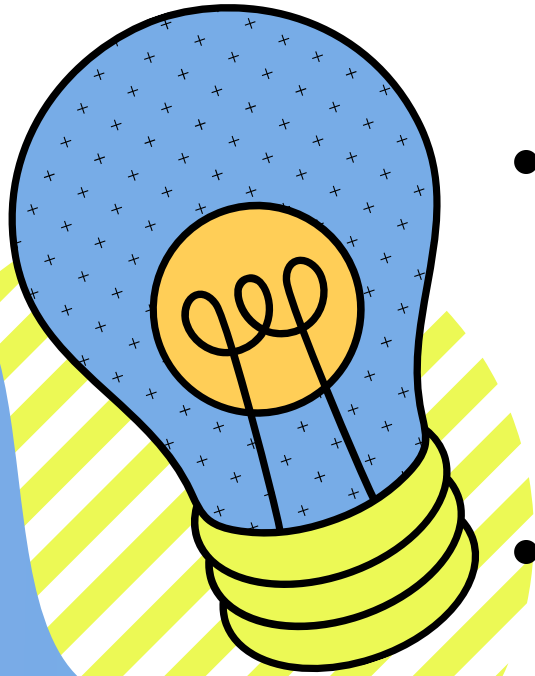
Modelling Philosophy

- How do we translate complex maths into C++ code?
- The 4-Step Process:
 - **Validation:** Using MATLAB to find the best signal "features"
 - **Extraction:** Setting up 100ms sliding windows for data capture.
 - **Translation:** Coding filters and RMS calculations in Arduino/C++.
 - **Actuation:** Mapping calculated values to motor movements.



Core Assumptions

- The Signal: usable EMG spectrum is between **10 Hz and 450 Hz**.
- The Relationship: linear correlation between muscle contraction intensity and the root mean square (RMS) of the voltage.
- Latency: 100ms window is the "sweet spot"

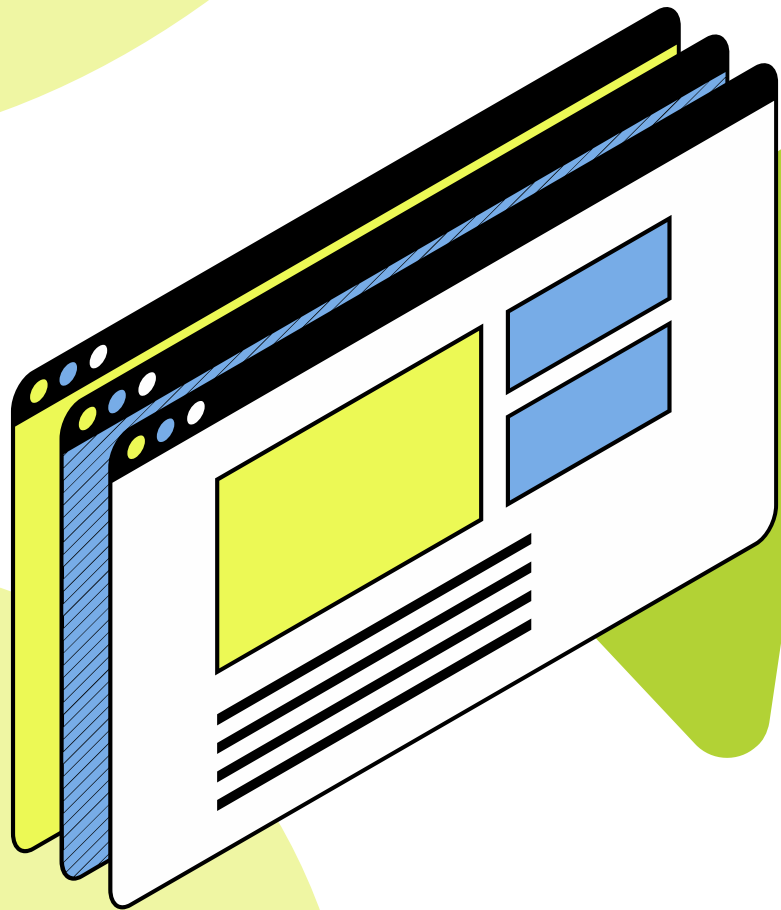


Data Inputs and Acquisitions



- Surface EMG sensors connected to the ESP32's 12-bit ADC.
- Sampling: High-frequency sampling (1kHz+)
- Preprocessing: Converting the AC signal to DC.
- Smoothing: Applying a Butterworth Bandpass filter to remove 50/60Hz powerline noise.

CORE: Mathematics



Why RMS?

- The most computationally efficient and physiologically meaningful indicator of muscle contraction intensity.
- The Logic: Instead of tracking every tiny spike in the raw signal, we calculate the average power over 100 ms.
- Application: This value determines the speed and strength of the prosthetic grip.

- Equation:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$$

Comparison RMS vs SSC

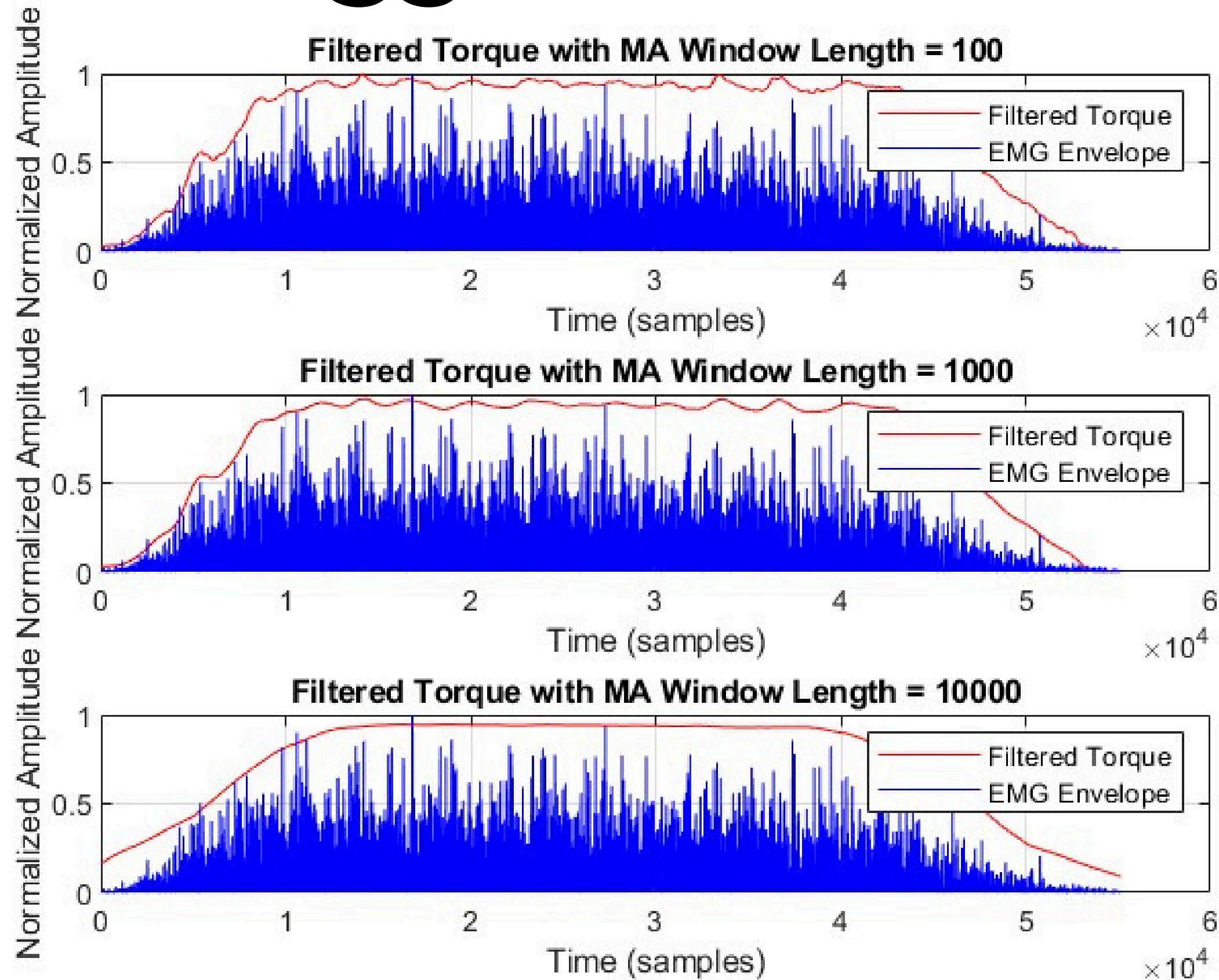
RMS

- Primary Function: Contraction Intensity.
- Used to distinguish between "Rest" and "Active" states. Strong force correlation makes it the backbone of movement strength control.

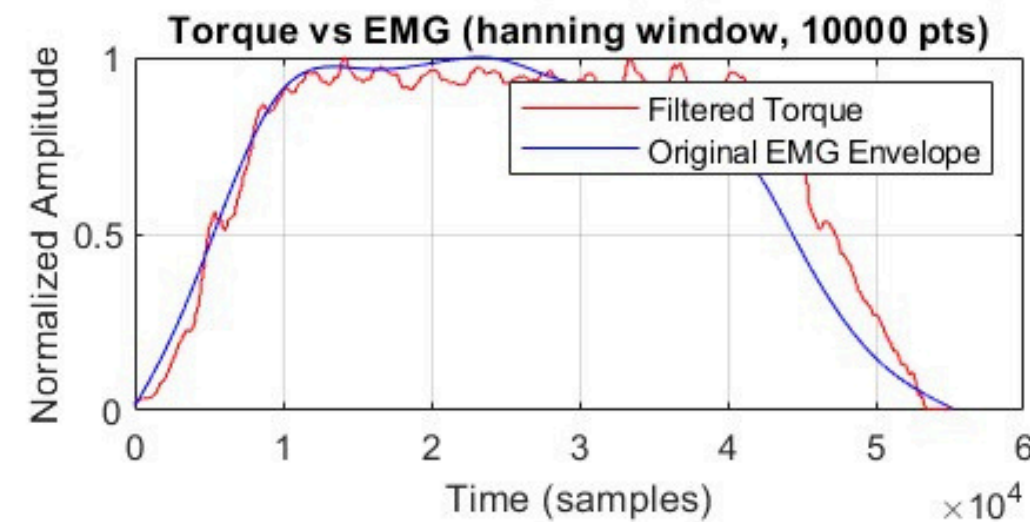
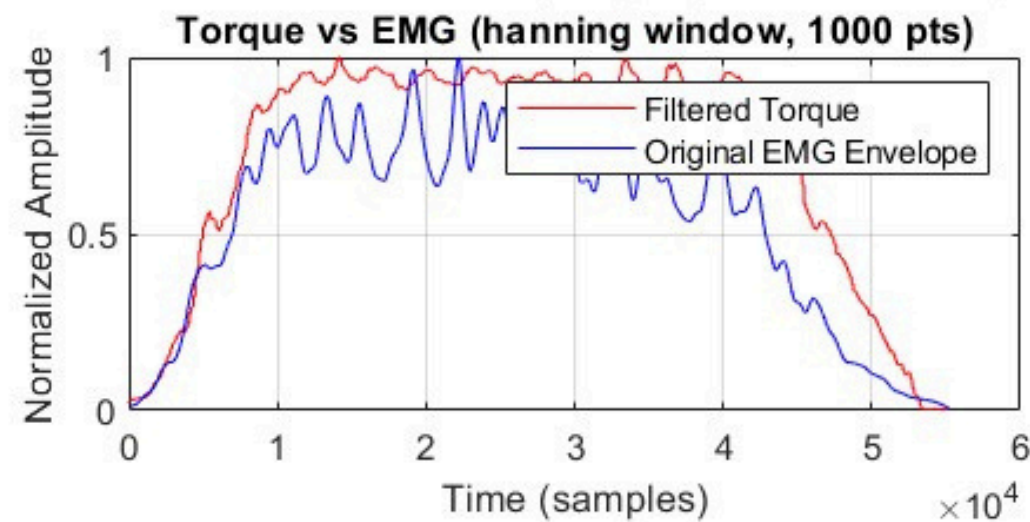
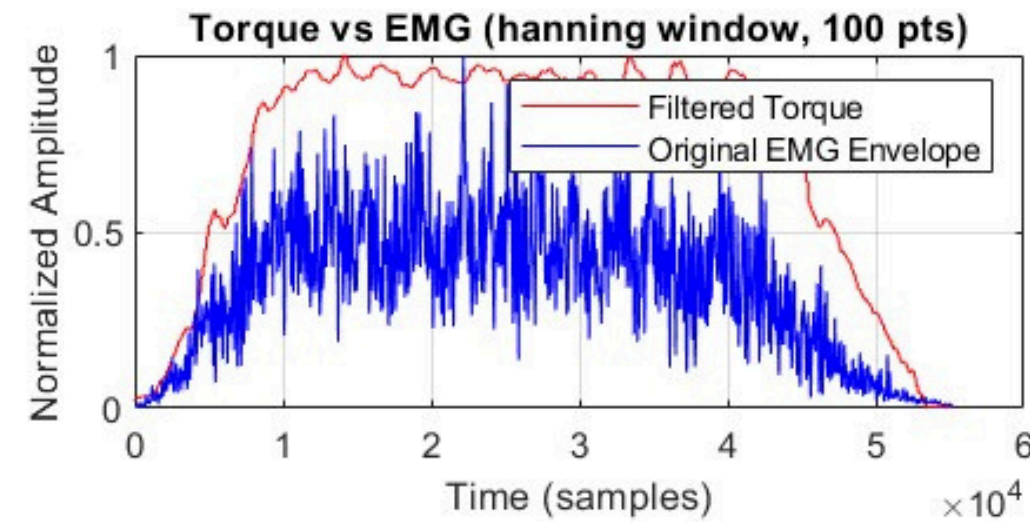
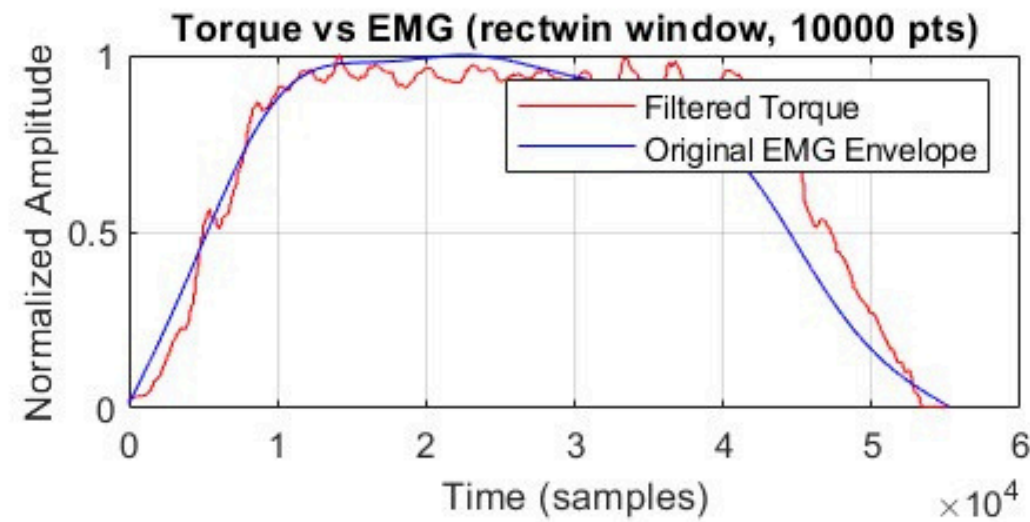
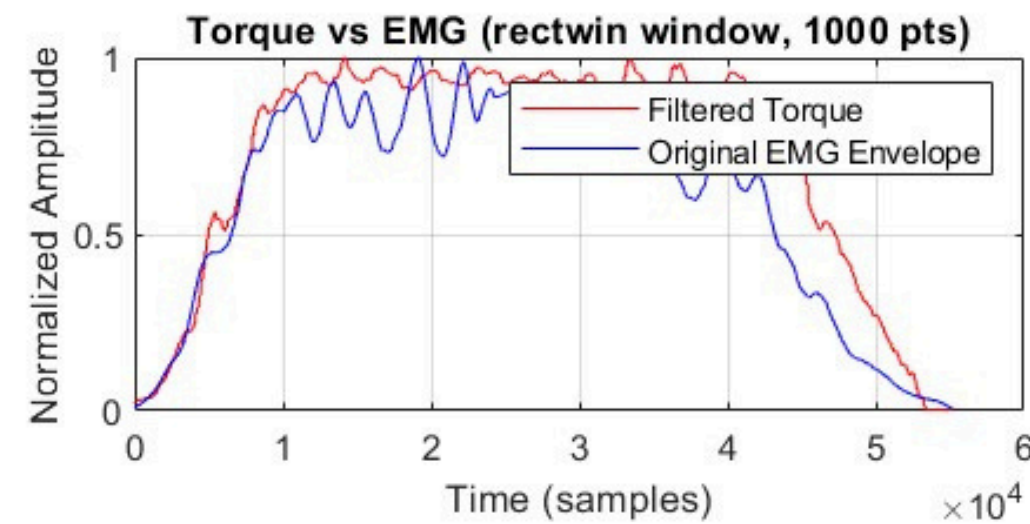
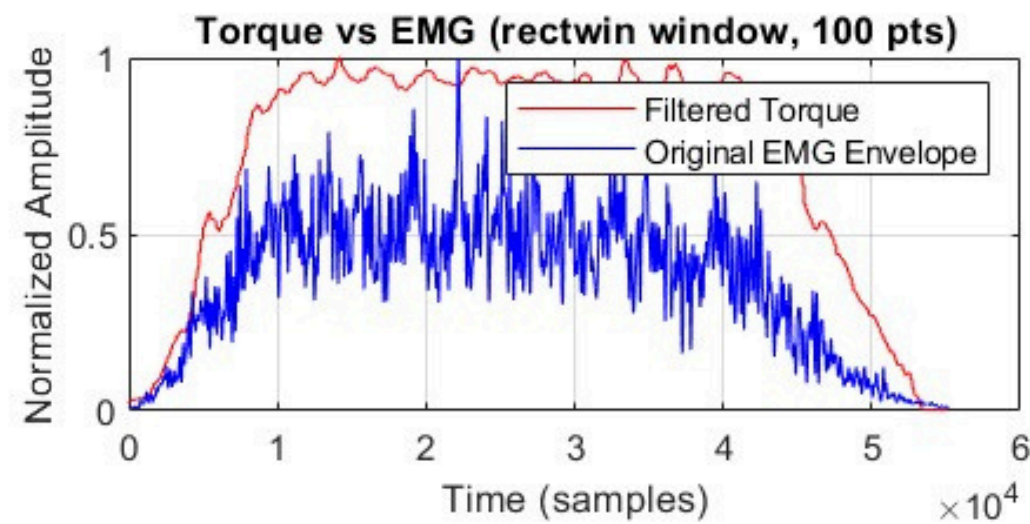
SSC

- Primary Function: Pattern Complexity.
- Essential for distinguishing "Fist" gestures from standard flexion/extension when both muscle groups show high RMS values.

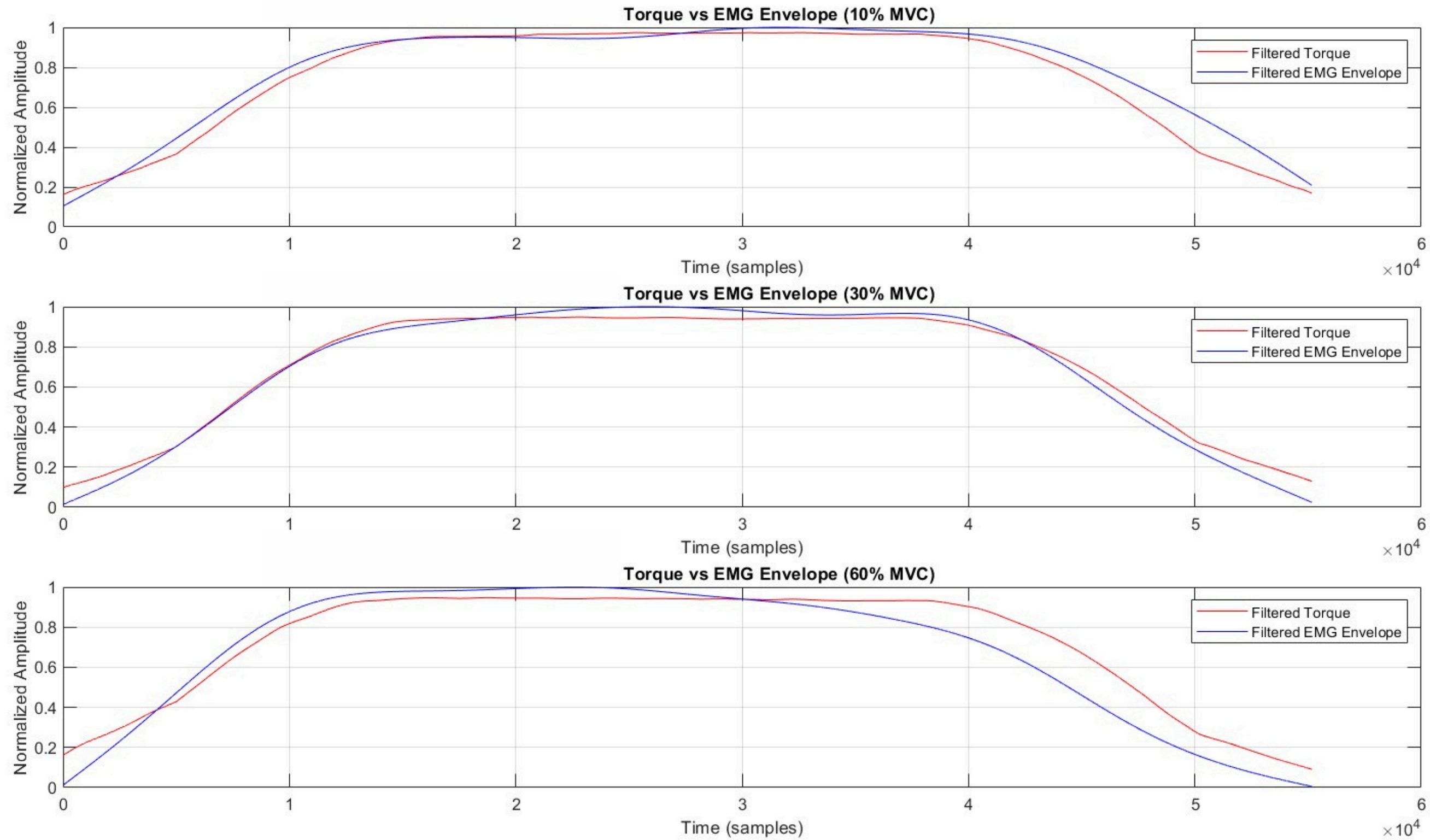
Methodology



Methodology



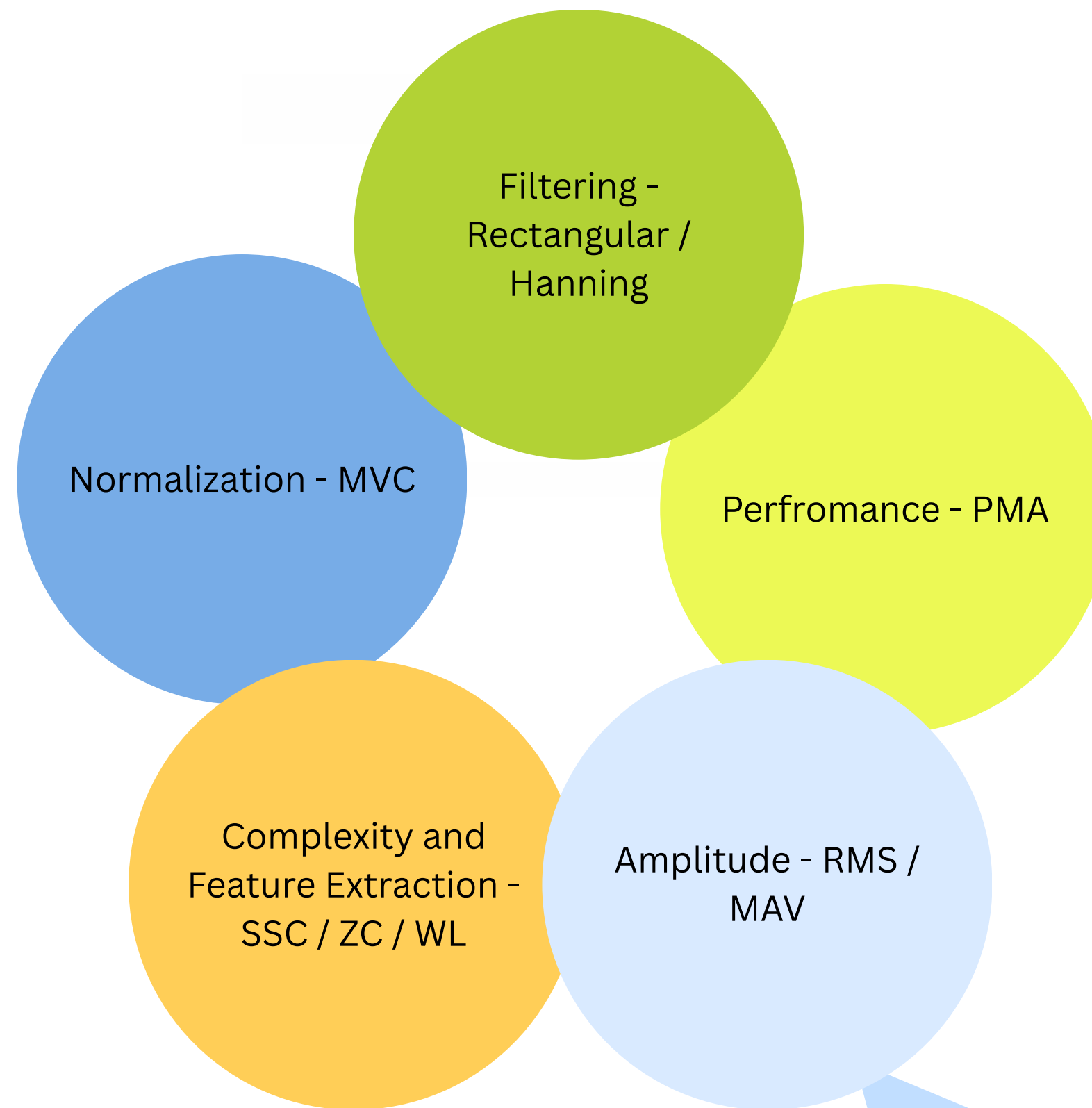
Methodology



Methodology

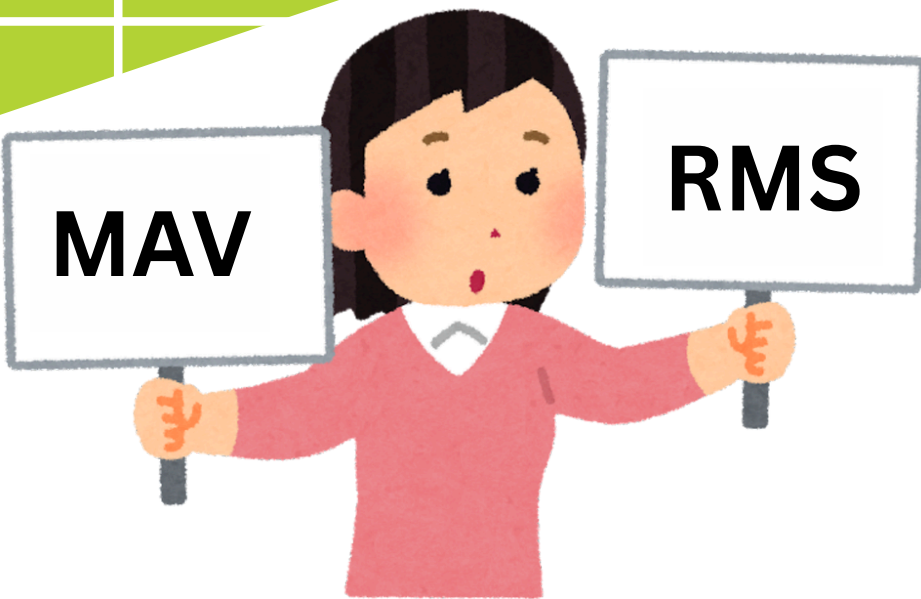
Term / Method	Full Form	Type	Main Purpose	How It Works	Output Meaning	Best Use Cases	Strengths	Limitations
MAV	Mean Absolute Value	Feature Extraction	Measure average muscle activation	Average of absolute EMG values over a window	Average signal magnitude	Real-time gesture control, prosthetics, rehab	Simple, fast, low computation	Sensitive to baseline noise
RMS	Root Mean Square	Feature Extraction	Measure signal power / effective activation	Squares → averages → square root	Signal energy/power	EMG intensity, fatigue, force estimation	Better reflects power than MAV	Slightly higher computation
MVC	Maximum Voluntary Contraction	Normalization / Calibration	Personalize EMG thresholds	Uses max voluntary contraction as baseline	% of personal maximum	Rehab progress, adaptive control	Standardizes across users/sessions	Affected by fatigue, effort inconsistency
PMA	Peak Mean Amplitude / Peak Muscle Activation	Performance Metric	Track highest activation reached	Peak feature value during session/task	Maximum observed activation	Session comparison, progress tracking	Useful for monitoring improvement	Definition may vary
ZC	Zero Crossing	Feature Extraction (Frequency-related)	Estimate signal frequency/activity changes	Counts how often signal crosses zero (with threshold)	Frequency of sign changes	Gesture classification, muscle state changes	Simple, useful for dynamic activity	Sensitive to noise without thresholding
SSC	Slope Sign Changes	Feature Extraction (Complexity)	Measure waveform complexity / muscle contraction changes	Counts slope direction changes between samples	Signal complexity/frequency behavior	Fine gesture recognition, movement transitions	Captures subtle dynamic changes	Noise-sensitive, threshold needed
WL	Waveform Length	Feature Extraction	Measure total waveform complexity and amplitude variation	Sum of absolute differences between consecutive samples	Combined amplitude + frequency information	EMG classification, motor control	Strong discriminative power	Can be affected by high-frequency noise
Rectangular Filter	Moving Average / Boxcar Window	Filter	Basic smoothing	Equal-weight average across window	Smoothed signal	Low-resource embedded systems	Very simple, low cost	Spectral leakage, weaker smoothing quality
Hanning (Hann) Filter	Weighted Window Filter	Filter	Improved smoothing with reduced edge effects	Weighted center-heavy averaging	Cleaner smoothed signal	EMG preprocessing, FFT, rehab control	Better frequency response	Slightly slower, more computation

Methodology-Logic



Methodology: Why RMS over MAV

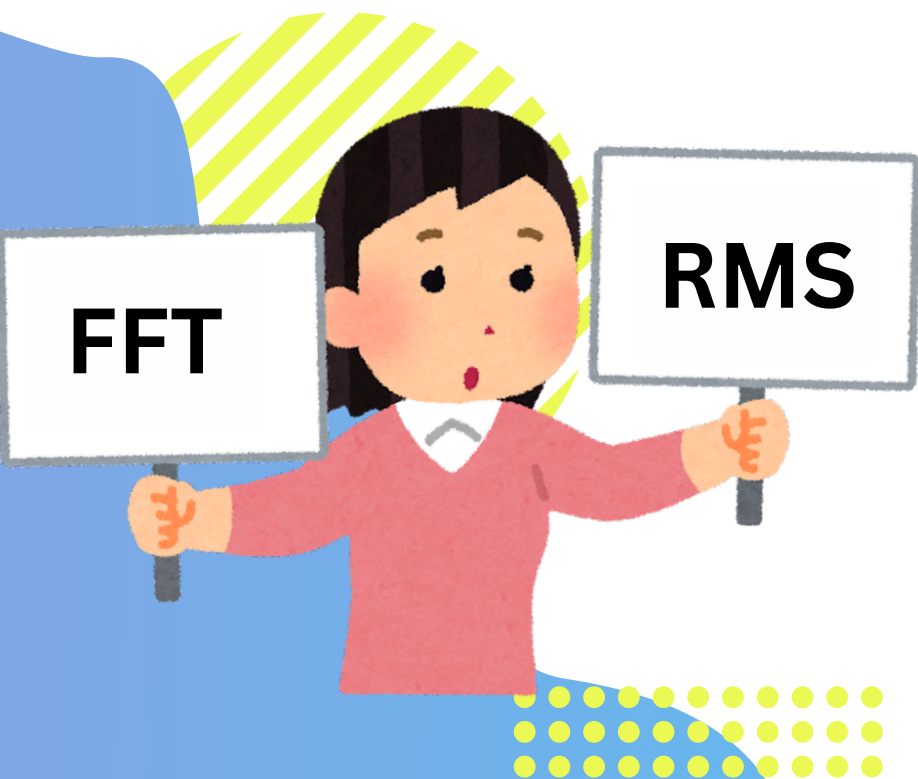
Decision & Discussion



Magnitude: signal power $\rightarrow$ physical force : RMS
Reason: Normalization

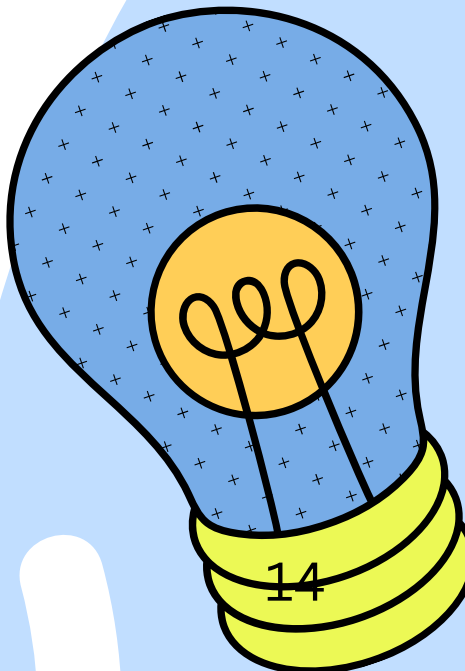


Rule-Based Classification $\rightarrow$ SSC + RMS $\gg$ RMS + M/L
Reason: Resource and Simplicity

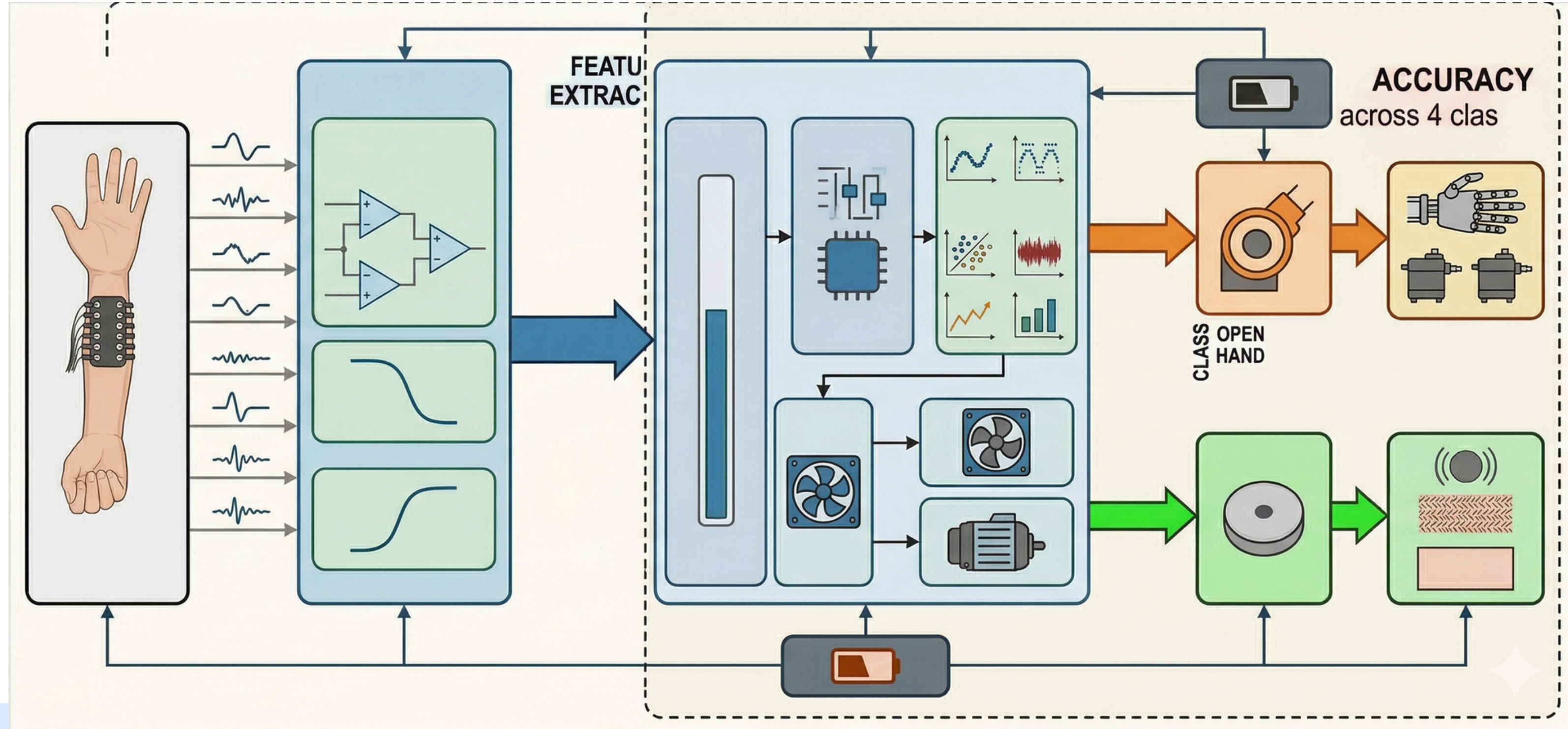


Real-Time Classification $\rightarrow$ RMS + SSC $\gg$ FFT
Reason: Timing and Resources

Structure: Classification



Flowchart



Preprocessing: 10-450Hz
Bandpass Filtering

Extraction: Sliding Window &
Feature Analysis

Classification: Rule-Based
Decision Logic

Actuation: Motor Output &
Haptic Pulse

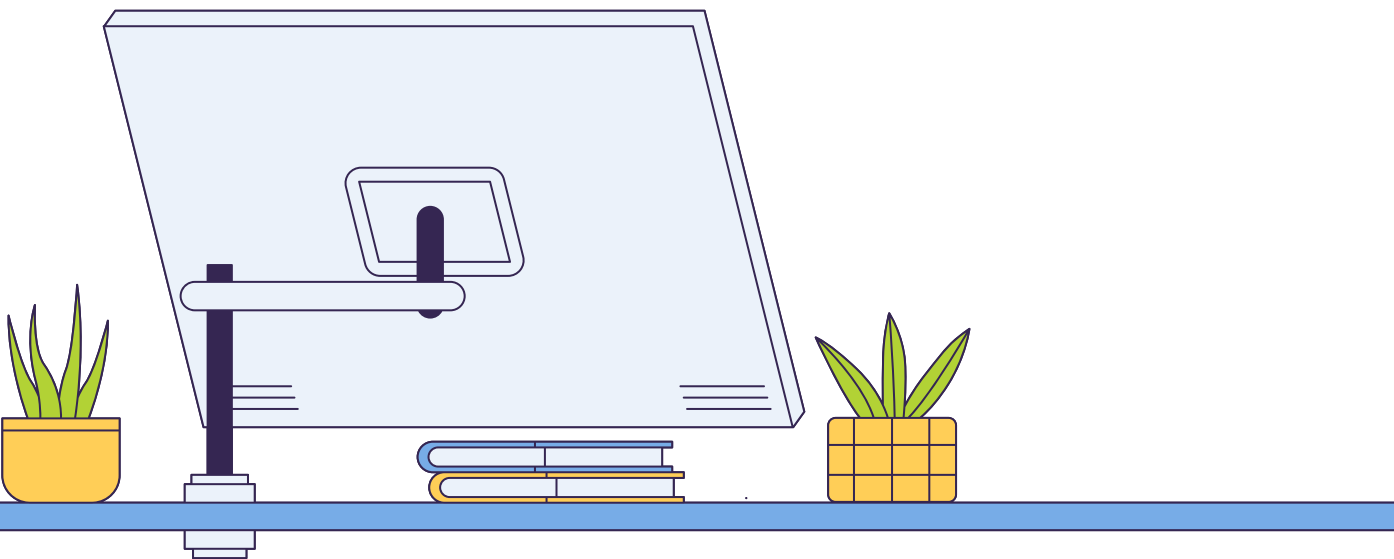
References & Resources

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Rule based Classification vs ML: Vujaklija, I., et al. (2017). "Online combinatorial control of robust EMG features."



Thank You

